

CSCI 5541 NLP HW3

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1 Introduction and Set up

This homework is about fine-tune a pretrained model to perform text classification. And I choose dataset Dataset: <https://huggingface.co/datasets/stanfordnlp/sst2>, and I choose DistilBERT to fine-tune. And I upgrade torch to 2.10.0+cu128 comply to my hardware. Furthermore, test accuracy was intentionally excluded from the analysis and visualizations, as the ground truth labels in the provided testing dataset were masked with dummy values -1.

2 Experiment

This homework asks us to do the experiment under 3 perspective, which are learning rate, epoch and training size.

2.1 Learning Rate

First, I training model us different learning rate, and result are the following:

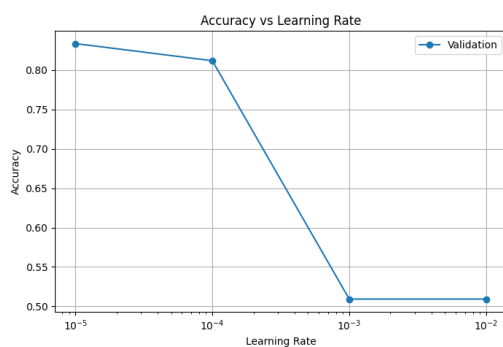


Figure 1: acc in different learning rate

I choose 4 different learning rates: $\{10^{-5}, 10^{-4}, 10^{-3}, 10^{-2}\}$ under 2 epoch, and training size is 800. We can see that 10^{-5} is the best number, but as the lr increases, the accuracy is decreases. I consider that its because the larger step size will break the structure between variables. We only need fine-tune here.

2.2 Epoch

The next part of the ablation experiment is tuning the epoch. I using epoch 1 to 10 in this experiment. Only 10 epochs just because the experiment is spend too much time. Specifically, I using train size 800 under different learning rate.

The first is to use the learning rate = 0.001

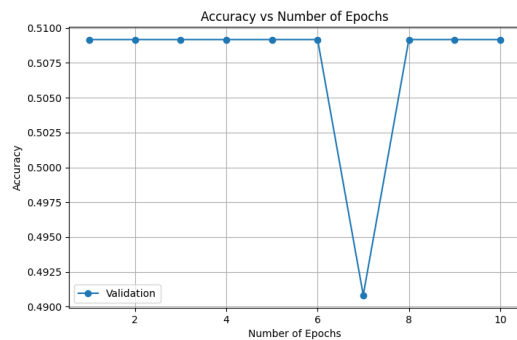


Figure 2: under learning rate 0.001

The Second is to use the learning rate = 0.0001

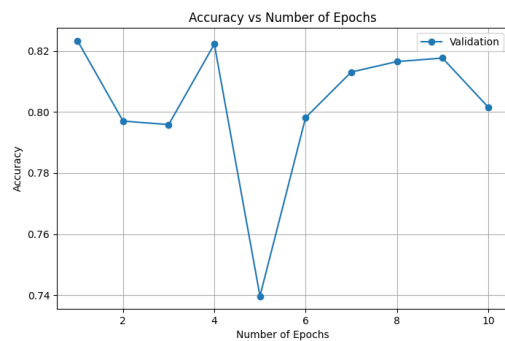


Figure 3: under learning rate 0.0001

The last is to use the learning rate = 0.00001

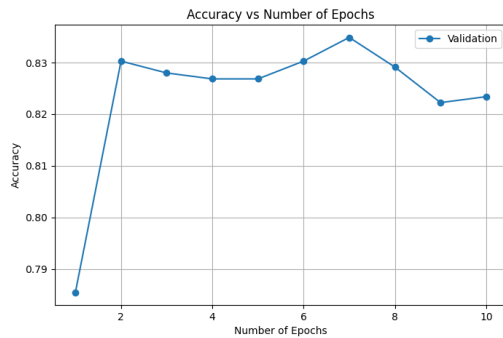


Figure 4: Enter Caption

On above 3 plot we can see the result of the different epoch at different learning rate: At 0.001, the model just guess. At 0.0001 the model is not robust and has lower accuracy. At 0.00001, the model is more stable than the previous one. As we can see, the epoch is not as high as better. I think that because the model will overfit when we train more on the part data.

2.3 training size

This part used different sizes of training data {50, 100, 400, 800} under 2 epoch with a learning rate 0.00001. The result is as follows:

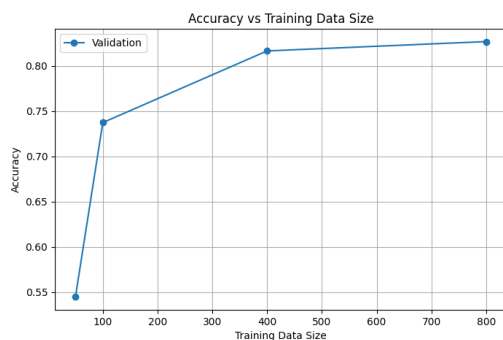


Figure 5: different training sizes

We can see that is training data. The size as much as better. I think that supplying more data will let the model to capture the different subtle features in fine tuning.

3 summary

The "Best Model" (utilizing a Learning Rate of 10^{-5} , 800 training samples, and 7 epochs) achieves optimal validation accuracy because it balances task adaptation with the preservation of DistilBERT's pre-trained knowledge. Specifically, the extremely low learning rate allowing the model to

gently fine-tune its weights without destroying its foundational language understanding. Simultaneously, maximizing the training size to 800 provides enough linguistic diversity for the model to learn robust sentiment patterns rather than memorizing noise. Finally, restricting the training to 7 epochs acts as an effective stopping point; this prevents the network from overfitting to the relatively dataset, ensuring the model generalizes effectively to unseen data.